# Jewels: Toward Text-Conditioned, Streamable, and Locally Editable Video as Space-Time Gaussian Fields

The Jewels Project

Technical report – August 5, 2026

**Status.** This is a progress report, not a benchmark or a claim of completed text-to-video generation. It separates established results, failed controls, and hypotheses still under test.

**Abstract**

We investigate whether ordinary RGB video can be generated and edited as an explicit field of anisotropic Gaussian primitives in normalized image-time coordinates rather than as a sequence of pixel frames. Each primitive, or *jewel*, occupies a volume in (u,v,t), carries opacity and a linear color field, and may tilt through time so that motion is represented by oriented space-time support. A deterministic additive renderer turns the set into frames. The longer-term model emits new jewels in overlapping time windows, retains persistent jewels by stable identity, and repairs only dirty latent cells after a user moves selected primitives.

This report consolidates the feasibility evidence to date. Synthetic tubes reconstruct at about 55 dB; four finalized 96-frame UCF-101 fields with 120k jewels replay at a mean 32.28 dB. A density audit shows that raw jewel count is misleading: doubling a UCF field from 45k to 90k isotropically increases effective per-frame contributors by only 30.3% and lowers full-volume PSNR by 1.07 dB, motivating temporal-preserving spatial densification. A local $32^3 \times 24$ tokenizer is the first held-out configuration to preserve small pedestrians; a train-only $2^3$ PCA hierarchy reduces it to $16^3 \times 96$ with a 0.224 dB penalty. An axial rectified-flow prior beats retrieval and mean-collapse distribution baselines but does not yet use its semantic condition. Conversely, a cell-local prefix model reduces future-birth feature error by 62.7% against a disjoint shuffled-prefix control and renders at 19.87 dB versus 7.30 dB. Edit planning and masked sampling preserve every clean latent exactly, although the repaired region remains blurry. A four-class prompt corpus passes its text-geometry preflight, while the prompt-conditioned generative experiment remains in progress.

Gaussian reconstruction of video is not new: VeGaS, GSVC, GaussianVideo, GaRV, and related work predate this project. The research opportunity evaluated here is narrower: a corpus-level generative distribution that directly emits persistent, editable 2D-video space-time Gaussian fields. Because the field is a representation rather than a sampling objective, it can in principle be generated by flow matching, diffusion, or next-token prediction. Its hierarchy also suggests a structured multimodal event language in which text, persistent visual state, and edit operations share one sequence. We state these as provisional systems hypotheses, not priority claims.

## 1 Introduction

Pixel video models repeatedly synthesize dense frames and must learn temporal coherence through architecture, loss, or data. Jewels asks whether coherence can instead be partly structural. We treat a video as a continuous three-dimensional signal over horizontal position, vertical position, and time. A moving visual element can then be represented by a Gaussian tube whose principal axis leans through time. The model is explicit: primitives can be selected, moved, recolored, carried between generation windows, culled at multiple levels of detail, and rendered at arbitrary times.

The intended system is not merely a codec. It is a text-to-video model whose native output is an editable space-time field. After generation, a viewer sees the video as a parallelepiped, drags a cursor through it, moves one or many jewels, and asks the model to resample only the vacated and destination neighborhoods. Untouched regions are hard constraints rather than soft conditioning.

The project has progressed through a sequence of falsifiable gates:

1. demonstrate that anisotropic space-time primitives can fit motion and ordinary video;
2. measure whether fitted sets are canonical enough for a shared generative model;

3. raise effective splat density without destroying temporal persistence;
4. learn a deterministic, count-aware local bottleneck for dense fields;
5. generate hierarchical latents with tractable attention;
6. prove exact edit locality and overlap-conditioned continuation; and
7. introduce text only after a leakage-safe prompt corpus exists.

Our contributions are therefore empirical and architectural rather than a completed generative benchmark. We report (i) an additive (u,v,t) Gaussian representation and gauge-free field features, (ii) density and tokenizer experiments that identify what preserves small moving subjects, (iii) controlled evidence that spatially aligned prefix context is essential for continuation, (iv) an edit/inpaint formulation with exact clean-region preservation, and (v) explicit negative results that bound what remains unproved. We additionally articulate two falsifiable systems hypotheses: that explicit persistent fields offer interaction properties unavailable from an opaque raster latent alone, and that a hierarchical field can serve as a multimodal visual event language.

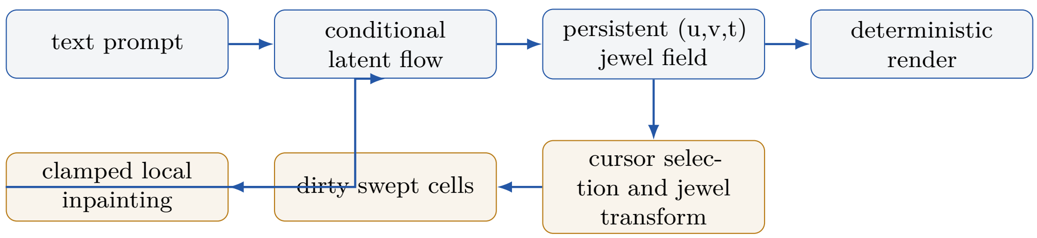


**Figure 1:** Target system. Generation emits an explicit field rather than pixels. Editing moves protected primitives, marks source, destination, and swept cells dirty, and resamples only those cells while the rest of the latent remains exact.

## 2 Relationship to Prior Work and Claim Scope

Modern Gaussian splatting was popularized for explicit 3D radiance fields by Kerbl et al. [5]. Dynamic-view methods subsequently added deformation, time, or native higher-dimensional primitives [7, 19]. Those systems reconstruct a view-dependent 3D scene; our target is the displayed 2D RGB video itself, without recovering physical depth or novel views.

That narrower video-representation lane is also established. Splatter a Video embeds monocular video into canonical 3D Gaussians with motion [16]. VeGaS uses folded Gaussian distributions to obtain time-conditioned 2D Gaussians and supports editing [14]. GSVC predicts frame-level 2D Gaussian sets from previous frames for compression [17]. GaussianVideo learns time-varying deformations of 2D Gaussians [6]. Most directly, GaRV encodes videos as 3D Gaussians in a space-time volume [13]. HyperGS amortizes reconstruction by predicting per-frame Gaussian representations feed-forward from input video [22]. Therefore, **we do not claim to be first to reconstruct 2D video with Gaussian splats, nor first to use a Gaussian space-time volume.**

Generative Gaussian work is substantial but usually targets 3D objects, rooms, or dynamic 3D scenes. GaussianCube and L3DG learn structured or latent generative models of 3D Gaussians [12, 20]. VideoRFSplat generates text-conditioned 3D scenes with a video model [3]; Diff4Splat predicts a deformable 3D Gaussian scene from an image, camera path, and optional text [10].

Our provisional distinction is the combination of: (1) direct generation of the displayed 2D video as an anisotropic (u,v,t) Gaussian field, (2) persistent stable-ID carry and birth emission across overlapping time windows, and (3) hard-local jewel manipulation followed by constrained latent repair. Our August 2026 search found no published system with this exact combination. However, absence from a search is not proof of priority, and the current project has not yet shown generalizing text-to-video generation. We therefore use "toward" in the title and reserve any first-of-kind claim for a completed, externally benchmarked model and a broader review.

**Table 1:** Positioning against the closest video-Gaussian families. "Amortized" means a shared encoder predicts a representation from input video; it is not the same as unconditional or text-conditioned generation from noise.

| Method | Native target | Temporal form | Shared prediction | Primary goal |
|---|---|---|---|---|
| VeGaS | 2D video | folded 3D Gaussians, conditional slices | no | representation/editing |
| GSVC | 2D video | linked per-frame 2D splats | no | compression |
| GaussianVideo | 2D video | time-deformed 2D splats | no | representation; codec |
| GaRV | 2D video | 3D space-time Gaussians | encoder/decoder scheme | random-access decoding |
| HyperGS | 2D video | predicted per-frame 2D splats | yes, from video | amortized encoding |
| JEWELS (target) | 2D video | persistent anisotropic (u,v,t) field | from noise and text | generation/editing |

# 3 Method

## 3.1 Video as an additive space-time field

Let the normalized video domain be $\Omega = [-1,1]^3$, with a query $\mathbf{x} = (u,v,t) \in \Omega$. Jewel $i$ has center $\boldsymbol{\mu}_i \in \mathbb{R}^3$, symmetric positive definite covariance $\boldsymbol{\Sigma}_i \in \mathbb{R}^{3\times3}$, weight logit $\ell_i$, base color $\mathbf{c}_i \in \mathbb{R}^3$, and a first-order color matrix $\mathbf{G}_i \in \mathbb{R}^{3\times3}$. Its contribution is

$$q_i(\mathbf{x}) = (\mathbf{x}-\boldsymbol{\mu}_i)^\top \boldsymbol{\Sigma}_i^{-1}(\mathbf{x}-\boldsymbol{\mu}_i), \tag{1}$$

$$a_i(\mathbf{x}) = \sigma(\ell_i)\exp[-q_i(\mathbf{x})/2], \tag{2}$$

$$\widehat{\mathbf{I}}(\mathbf{x}) = \mathbf{b} + \sum_i a_i(\mathbf{x})\left[\mathbf{c}_i + \mathbf{G}_i(\mathbf{x}-\boldsymbol{\mu}_i)\right], \tag{3}$$

where $\mathbf{b}$ is a learned constant background. This is additive accumulation, not ordered alpha compositing. The P1 color term avoids requiring many constant-color primitives to approximate smooth gradients.

Writing $\boldsymbol{\Sigma}_i = \mathbf{R}_i \operatorname{diag}(\mathbf{s}_i^2)\mathbf{R}_i^\top$ makes the motion interpretation explicit. If a principal axis of $\mathbf{R}_i$ tilts into time, the support is a sheared tube through the volume. Rendering a frame means querying the plane at its $t$ coordinate. The representation therefore has continuous random access in space and time.

Training samples arbitrary voxels, minimizes RGB MSE, and periodically splits high-gradient primitives while pruning low-weight ones. Euclidean-center kNN supplies candidate splats; exact Mahalanobis weights are then evaluated for those candidates. Distance work is chunked to at most 100M point-center pairs, allowing 45k–120k-field experiments on an 8 GB RTX 2070 Super. Atomic checkpoints include optimizer and random-generator state, giving exact continuation across process failure and CPU/CUDA densification boundaries.

## 3.2 Gauge-free features and deterministic tokenization

Raw scale/quaternion parameters are unsuitable learning targets: quaternions double-cover rotation and covariance axes may permute. We instead use the unique log-Euclidean coordinate

$$\log \boldsymbol{\Sigma}_i = \mathbf{R}_i \operatorname{diag}(2\log \mathbf{s}_i)\mathbf{R}_i^\top. \tag{4}$$

The upper triangle contributes six values to a 22-D feature: $[\boldsymbol{\mu}, \operatorname{triu}(\log \boldsymbol{\Sigma}), \mathbf{c}, \operatorname{vec}(\mathbf{G}), \ell]$. Decoding by eigendecomposition reconstructs a render-equivalent field with approximately $10^{-6}$ maximum feature round-trip error.

Dense fields are packed into a fixed $32^3$ cell raster. The count-aware local encoder pools means, variances, counts, occupancy, and canonical rank features; the sparse decoder emits only the predicted variable number of jewels rather than allocating worst-case slots for every cell. A train-only PCA over non-overlapping $2^3$ blocks maps eight 24-D fine cells into one 96-D coarse code. The resulting $16^3$ grid is modeled with rotating axial attention: one block attends along $u$, the next $v$, and the next $t$. Three blocks communicate globally without materializing a $4096^2$ attention matrix.

The full-field prior uses rectified flow [8]. For clean target latent $z_1$, noise $z_0$, and interpolation $z_s = (1-s)z_0 + sz_1$, the model regresses the velocity $z_1 - z_0$ while receiving time and an optional CLIP condition [11]. Classifier-free dropout is required before the null branch is interpreted.

### 3.3 Persistent continuation

Long video is not generated by concatenating independent fixed clips. A stride retains jewels whose finite support intersects the overlap and copies their stable IDs and parameters exactly. The model predicts only new birth counts and 22-D birth marks in frontier-local coordinates. A $16 \times 16 \times 8$ raster of prefix counts and moments supplies spatially aligned context tokens. This factorization turns “splat density per unit time” into an explicit birth process: the field maintains a target contributor density while old jewels persist and new regions receive births.

### 3.4 Cursor edits and constrained repair

A cursor selects jewel centers or support intersections in the parallelepiped. Translating a group creates protected moved jewels and a dirty mask covering the source, destination, swept axis-aligned bounding box, and a cell halo. Let $m \in \{0,1\}^K$ mark dirty latent cells and $z^{\text{clean}}$ be the original hierarchy. After each Euler flow step, sampling reclamps

$$z_{k+1} \leftarrow (1-m) \odot z^{\text{clean}} + m \odot \left[z_k + \Delta s\, v_\theta(z_k, s_k, c, m)\right]. \tag{5}$$

The final merge contains untouched jewels, generated jewels decoded from dirty cells, and protected moved jewels. This guarantees locality mechanically; visual reconciliation at the destination must still be learned from edit tuples.

## 4 Experimental Protocol

Experiments use three tiers. A synthetic moving tube falsifies the basic motion representation. CUHK Avenue [9] supplies fixed-camera fitting and generation-mechanics experiments. UCF-101 [15] supplies diverse action and appearance tests. Unless stated otherwise, the short side is 160 pixels. The current prompt corpus uses 96-frame, 120k-jewel joint fits with temporal-preserving spatial splitting.

We intentionally distinguish metrics. Fit-training PSNR is often computed on sampled voxels and is not a publication-grade codec metric. “Full-volume PSNR” replays the complete saved field against the decoded source. Tokenizer PSNR compares continuous samples rendered from the fitted target and the decoded target, not decoded output against the source pixels. Energy distance evaluates latent sample distributions. Correct/shuffled/null continuation controls use identical targets and model weights. These quantities answer feasibility questions and should not be compared directly with published rate-distortion or FVD results.

## 5 Results and Discoveries

### 5.1 Reconstruction is viable, but not novel by itself

The synthetic anisotropic-tube test reaches about 55 dB. A real fixed-camera clip reaches 43.2 dB with 10k primitives; four recent UCF prompt-corpus fields reach 30.32–33.67 dB on exact checkpoint replay. Figure 2 shows the source, 120k-jewel reconstruction, and five-times-amplified absolute error for basketball. Class identity, player silhouettes, court geometry, and dominant colors survive. Error concentrates on thin limbs, the ball, faces, object edges, and motion boundaries.

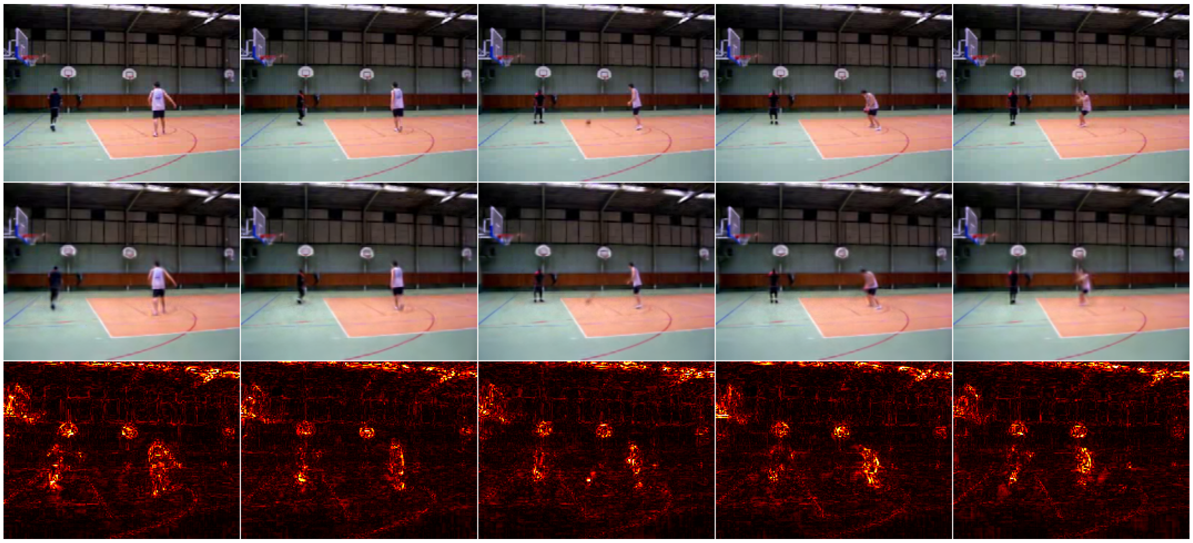

**Figure 2:** Exact 96-frame checkpoint replay sampled at frames 0, 24, 48, 72, and 95. Rows are source, 120k-jewel reconstruction, and 5× mean absolute RGB error. Full replay PSNR is 33.67 dB.

The covariance is not an ornamental parameter. Median anisotropy is about 2.8–3.3 on real footage, versus about 1.2 on the synthetic background-dominated test. More importantly, attempts to impose a soft Voronoi partition reduced both reconstruction and cross-seed canonicality. On one real clip, additive splats score 43.2 dB with a 0.621 normalized cross-seed Chamfer ratio; vanilla Voronoi scores 36.8 dB/0.745. A background pseudo-cell recovers some PSNR but not canonicality; Lloyd relaxation collapses to 33.0 dB/0.784. Equalizing cell sizes fights the desired content-adaptive allocation, so the branch was removed.

## 5.2 Useful density is a rate, not a raw count

At 45k total jewels, temporal $3\sigma$ support intersects roughly 6.2k splats per Avenue frame, but only 3.9k exceed 5% potential peak alpha and the opacity-weighted effective count is 3.3k. UCF is lower: 3.6k and 2.9k. Thus a nominally dense field remains below the desired 5k–10k effective contributors per frame.

The naive 90k isotropic control demonstrates why. It nearly doubles observed births per frame, shortens median support from 5 to 3 frames, increases effective contributors only 30.3%, and lowers PSNR. Figure 3 shows that extra count does not recover more useful detail.

**Table 2:** Matched UCF density control. The successful next control preserves temporal extent and splits spatial axes rather than shrinking every axis.

| Measurement | 45k isotropic | 90k isotropic | Change |
|---|---|---|---|
| Peak alpha $\geq$ 5%/frame | 3,595 | 4,724 | +31.4% |
| Effective contributors/frame | 2,933 | 3,822 | +30.3% |
| Full-volume PSNR | 34.136 | 33.062 | −1.074 dB |
| Mean support lifespan (frames) | 8.41 | 5.58 | −33.6% |
| Median support lifespan (frames) | 5 | 3 | −40.0% |
| Observed births/frame | 659 | 1,290 | +95.6% |
| Fit runtime (minutes) | 56.8 | 79.7 | +40.3% |


**Figure 3:** Source, 45k fit, and 90k isotropic fit at the same UCF frame. Doubling count increases short-lived fragments rather than useful detail and reduces full-volume PSNR.

## 5.3 A generative field prior works before it works well

An early permutation-invariant SetDiT flow model directly generated 6,471-jewel Avenue fields from noise. Scaling 5.8M, 58M, and 173M-parameter models under a frozen protocol monotonically improves CLIP-Frechet distance from 0.31 to 0.20 to 0.16 and training loss from 1.148 to 1.066 to 0.993 (Figures 4 and 5). Samples preserve scene layout, palette, chirality flips induced by mirror augmentation, and temporal stability, but fine pedestrians remain distorted. The flattening curve identified data and representation density, rather than further model scaling on 231 windows, as the next axis.

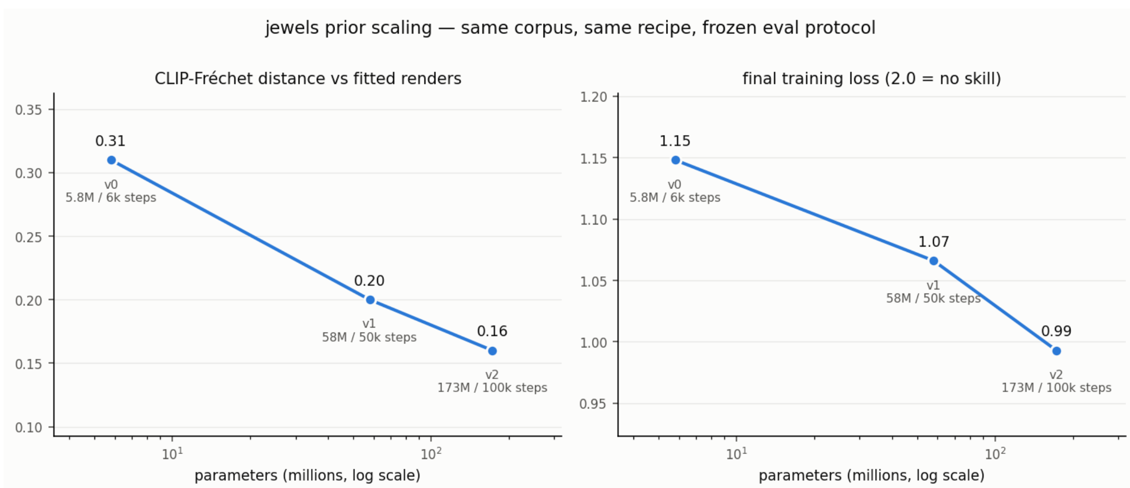


**Figure 4:** Frozen scaling protocol for the first direct field prior. Improvement is monotone but saturates on the small, single-scene corpus.

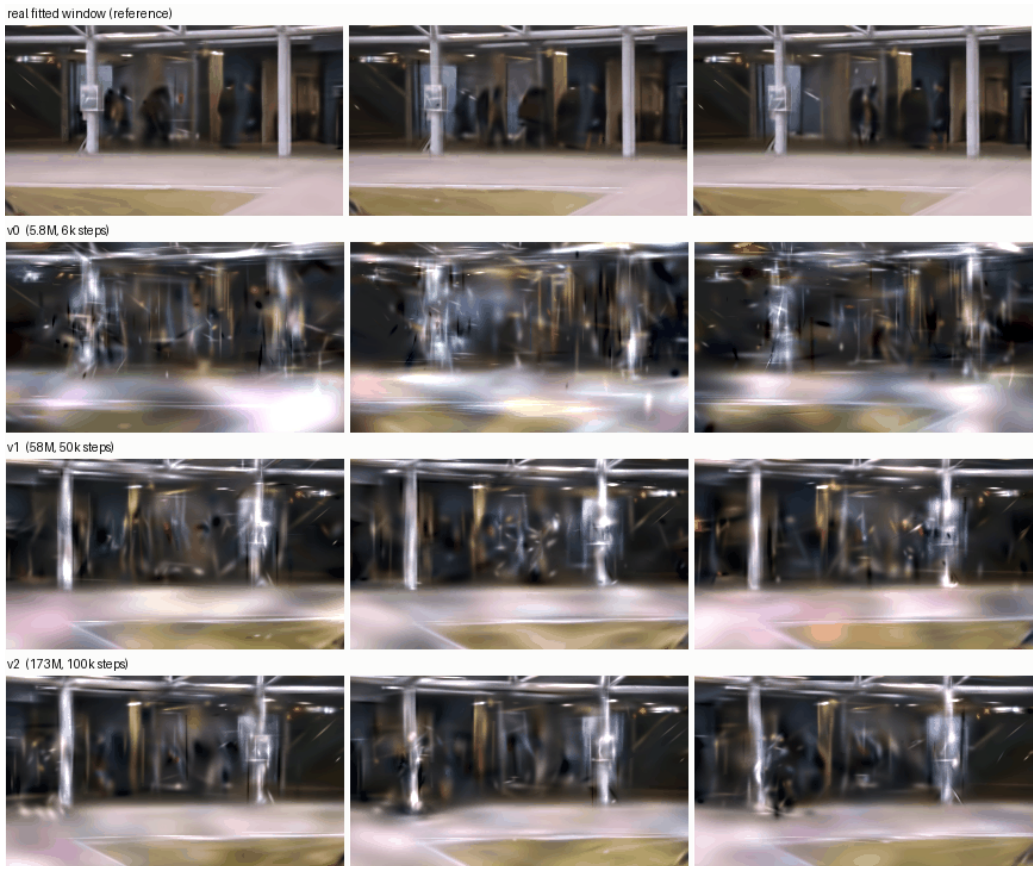


**Figure 5:** A fitted reference and samples from v0, v1, and v2 at three times. Larger models preserve the coarse scene more consistently, while people remain visibly under-resolved.

### 5.4 Locality is load-bearing for the tokenizer

On 121 dense Avenue windows, coarse cell pooling repeatedly achieved plausible aggregate PSNR while erasing people. The first source-held-out tokenizer to preserve recognizable pedestrian silhouettes uses a $32^3$ grid, 24 values per cell, near-one-jewel average locality, and canonical-rank-aware decoding. Its 35-window audit reaches 19.987 dB and 99.14% count recovery. This remains a weak 1.26× numeric bottleneck. The $2^3$ PCA hierarchy reduces 32,768 tokens to 4,096 and loses only 0.224 dB, making axial generation tractable.

Appearance is not solved. A bright red coat in held-out footage retains its figure but decodes tan. Rare foreground chroma is overwhelmed by uniformly weighted 22-D jewel losses and sparse uniform render samples. This is a regression-to-palette failure, not a channel-order bug. Foreground chroma and perceptual metrics must accompany aggregate RGB PSNR.

Cross-domain controls sharpen the conclusion. Frozen Avenue weights reach only 17.786 dB/86.06% count on UCF basketball; the identical architecture trained on that UCF window reaches 22.316 dB/97.13%. Equal-exposure joint training reaches 20.278 dB on held-out Avenue and 22.275 dB on UCF. The bottleneck can represent both, but absolute cell/domain specialization remains substantial.

**Table 3:** Selected tokenizer and hierarchy results. PSNR compares fitted target and tokenizer round-trip renders at fixed continuous samples.

| Representation | Tokens | PSNR | Visual result |
|---|---|---|---|
| $12 \times 12 \times 6$, 128-D | 864 | 18.940 | people erased |
| $16 \times 16 \times 8$, rank-v2 | 2,048 | 20.428 | people erased |
| $24 \times 24 \times 16$, pooled | 9,216 | 19.730 | faint traces |
| $32^3$, rank-aware, 24-D | 32,768 | **19.987** | people recognizable |
| $16^3$, PCA, 96-D | 4,096 | 19.763 | people retained |

## 5.5 Generation, continuation, and editing separate into distinct gates

The 1.94M-parameter axial prior learns the coarse latent distribution: energy distance 0.2493 beats CLIP nearest-neighbor retrieval (0.3164) and repeated scene mean (0.8473). Yet correct CLIP conditioning is slightly worse than shuffled or unconditional conditioning (gain $-0.00171$). With only four training cameras and one image embedding per window, the model learns scene-family statistics, not semantic control.

Continuation gives a stronger causal control. Replacing one global prefix vector with aligned $16 \times 16 \times 8$ local tokens produces the results in Table 4 and Figure 6. Correct context reduces standardized birth-feature MSE by 62.7% against disjoint shuffled context and recovers 99.95% of births. Carried jewels are copied with zero error. This is still an overfit to four views of one clip, but it proves that spatial correspondence, not merely window identity, is required.

**Table 4:** Matched cell-local continuation control.

| Metric | Correct | Shuffled | Null |
|---|---|---|---|
| Birth-feature MSE | **0.4475** | 1.1988 | 2.1695 |
| Birth-count MAE/cell | **0.3310** | 3.7167 | 6.4994 |
| Future-field PSNR | **19.870** | 7.302 | $-29.735$ |

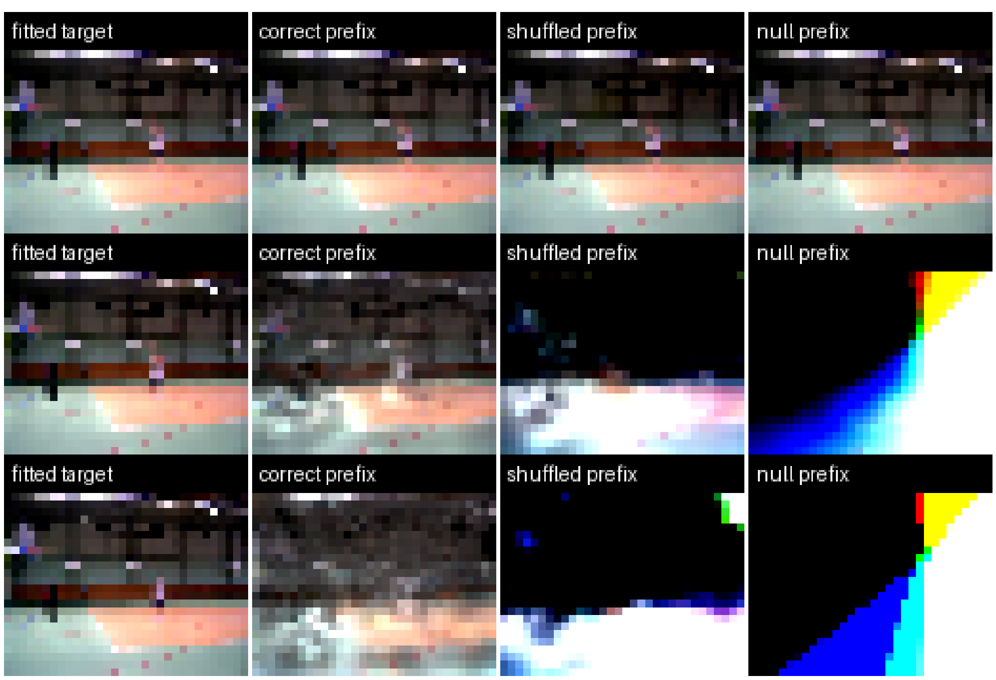


**Figure 6:** Start, middle, and end diagnostics for fitted target, correct prefix, disjoint shuffled prefix, and null prefix. The extreme null output is not a video baseline because this overfit was not trained with context dropout.

The editor translates 281 real fitted jewels, dirties 448 of 4,096 coarse cells (10.94%), and merges 45,244 final jewels. Every clean coarse and fine latent is bit-identical. The repaired strip, however, is visibly washed out. A mask-aware 5,000-step branch improves dirty latent MSE only from 1.8894 to 1.8848 and does not improve the visual. The result validates the editor invariant while falsifying the current repair learner.

### 5.6 Text preflight passes; promptable generation remains open

The prompt smoke test selects Basketball, HorseRiding, PlayingGuitar, and ApplyEyeMakeup from four disjoint UCF source groups: 12 training videos and four group-held-out validation videos. Three training phrasings and one unseen evaluation phrasing per class are embedded with frozen OpenCLIP ViT-B/32. Each unseen phrase retrieves the correct training-phrase centroid; the minimum correct versus best-wrong cosine margin is 0.2395 (Table 5).

**Table 5:** Held-out text geometry before jewel-model training.

| Class | Correct cosine | Best wrong | Margin |
|---|---|---|---|
| Basketball | 0.8641 | 0.5881 | 0.2760 |
| HorseRiding | 0.9004 | 0.6610 | 0.2395 |
| PlayingGuitar | 0.9037 | 0.6546 | 0.2490 |
| ApplyEyeMakeup | 0.9102 | 0.6229 | 0.2873 |

Four of 16 dense 120k-jewel fits are finalized, with replay PSNRs 32.10, 33.67, 30.32, and 33.15 dB (mean 32.28). The remaining fits are in progress. The decisive future test is not text retrieval; it is whether correct prompts beat shuffled prompts and a trained null condition on held-out source group 4 while preserving rollout density, color, identity, and motion.

## 6 What Has Been Falsified

Failures have been used as architectural evidence rather than hidden tuning history:

**Table 6:** Selected falsifications and the resulting decision.

| Failed hypothesis | Decision |
|---|---|
| Soft Voronoi improves canonicality | Removed; additive wins reconstruction and normalized cross-seed Chamfer. |
| More raw splats automatically increase useful density | Rejected; audit contributor rate and preserve temporal axes during split. |
| Aggregate PSNR selects a visually useful tokenizer | Rejected; require held-out motion silhouettes and foreground chroma checks. |
| Global prefix context is sufficient for births | Rejected; aligned cell-local context improves future-field PSNR by 3.41 dB. |
| The six-camera CLIP prior uses semantics | Rejected; correct condition does not beat shuffled/null. |
| Mask visibility alone teaches repair | Rejected; train explicit edit tuples with protected-jewel summaries. |
| Further model scaling solves the sparse prior | Deprioritized; the curve saturates on 231 single-scene windows. |

## 7 Limitations

- **No completed text-to-video result.** The four-class text geometry is valid, but the prompt-conditioned generator has not been trained or evaluated.
- **Small and specialized corpora.** The generative results use one fixed-camera scene; the continuation result is an overfit; the prompt corpus contains only 16 videos.
- **Fixed camera and no cuts.** Global camera motion shears most primitives at once and spends density on camera motion. Camera tokens or explicit stabilization are future work.

- **Low resolution.** Current fits use a 160-pixel short side. PSNR values are not comparable to full-resolution codec papers without a matched rate-distortion protocol.
- **Appearance loss.** Uniform feature and RGB losses underweight rare foreground color, faces, hands, and thin objects. The red-to-tan coat error is a concrete warning.
- **Pure-PyTorch rendering.** Exact kNN/Mahalanobis evaluation is useful for research but slower than a tile CUDA rasterizer. Current reconstruction is optimization-based, not amortized.
- **Edit quality.** Clean-region identity is exact, but dirty-region synthesis is not yet visually useful and may create additive collisions with protected moved jewels.
- **Novelty is provisional.** Closely related literature is moving quickly. A formal claim requires a complete system, broader search, matched baselines, and external review.

# 8 Why an Explicit Jewel Field Rather Than a Raster Latent?

The relevant comparison is not "jewels versus diffusion." A jewel field specifies the state that is represented and rendered; diffusion, flow matching, and autoregression specify how a model learns or samples a distribution over states. Video diffusion models usually operate on dense pixel or learned spatiotemporal latent grids [1, 4]. The rectified-flow prior used here is itself closely related: flow matching supports Gaussian probability paths that include diffusion paths as special cases [8]. A future JEWELS model could therefore use a diffusion or flow objective while retaining an explicit field as its decoded state.

Dense raster latents have formidable advantages: regular tensors map efficiently to accelerators, convolutions and attention are mature, topology changes are implicit, and large pretrained image and video priors already exist. The case for jewels is instead a set of interaction and state properties summarized in Table 7. These properties do not by themselves imply better photorealism or rate-distortion.

**Table 7:** Representation-level hypotheses relative to an opaque raster video latent.

| Field property | Potential capability | Required evidence |
|---|---|---|
| Stable primitive identity | Exact carry across generation windows | Held-out autonomous rollouts without identity drift |
| Localized finite support | Hard-clamped local repair after direct manipulation | Perceptually convincing edits with unchanged clean regions |
| Adaptive primitive density | Spend capacity on motion, boundaries, and rare detail | Matched quality and rate at 5k–10k effective contributors/frame |
| Continuous (u,v,t) domain | Random-access time slices and resolution-independent rendering | Quality and speed against matched video decoders |
| Explicit marks and covariance | Select, move, recolor, split, or delete native state | User edits that do not require global video regeneration |

The present motion noise is therefore a diagnostic, not something raw scaling is assumed to erase. The 45k-to-90k isotropic control increased effective contributors by only 30.3%, shortened support lifetimes, nearly doubled births, and reduced PSNR. More jewels should sharpen silhouettes only when densification splits spatial axes while preserving temporal support, and when motion-, edge-, and foreground-weighted samples prevent static background from dominating the objective. More fitting steps can let new primitives settle, while a shared generative prior may suppress inconsistent fit noise; neither can restore detail discarded by the representation or tokenizer. The direction should be rejected if matched temporal-preserving density fails to reduce motion-boundary error, if the tokenizer approaches one token per jewel, or if local edits still require near-global repair.

# 9 Jewel Fields as a Multimodal Event Language

Native multimodal generation is established prior art. Chameleon models arbitrary interleaved image and text token sequences [2]; Emu3 applies next-token prediction to discrete image, text, and video tokens [18]; and Transfusion combines language modeling and continuous image diffusion in one transformer [21]. We therefore do not claim that an LLM generating visual media is new. The narrower opportunity is to make the generated visual tokens persistent, spatially executable state rather than only codes that decode to a raster.

A jewel video admits an event factorization such as

$$\texttt{WINDOW}, \quad \texttt{CARRY}(i), \quad \texttt{CELL}(c), \quad \texttt{BIRTH}(n), \quad \texttt{MARK}(\theta), \quad \texttt{MOVE}, \texttt{SPLIT}, \texttt{DELETE}, \tag{6}$$

where $i$ is a stable primitive identity, $c$ is a space-time cell, $n$ is a local birth count, and $\theta$ contains continuous position, covariance, opacity, and color marks. Text, audio events, camera instructions, selections, and edit commands could be interleaved with this stream. The model output would then be both media and an executable edit program: retained state is copied exactly, new events extend the frontier, and an edit dirties only the affected cells.

A flat sequence of 120k jewels per clip would be impractical and would discard spatial locality. The selected hierarchy suggests a tractable factorization: autoregress or mask-model the $16^3$ coarse cell tokens and discrete event/count symbols, then use a local continuous head or conditional flow to emit fine 22-D jewel marks inside occupied cells. This also avoids pretending that every jewel is already an object-level semantic token; current primitives are low-level basis functions, and stable IDs are not automatically object identities. The multimodal-language hypothesis succeeds only if the hierarchy achieves substantial token compression, text controls held-out motion rather than retrieving a training clip, persistent events survive long rollout, and the same structured state supports cheaper local editing than full latent regeneration.

# 10 Path to a Promptable Model

The shortest evidence-driven path is:

1. complete all 16 leakage-safe 96-frame UCF fits and audit effective contributor density;
2. train one shared tokenizer/normalizer using train groups only, adding foreground chroma and motion-weighted render samples without sacrificing exact count contracts;
3. train initial-window and cell-local birth models with text dropout, using the same text embedding at every stride and explicit correct/shuffled/null controls;
4. evaluate one-step held-out continuation before autonomous rollout, then measure seams, birth rate, identity, color, and density across multiple generated strides;
5. create edit tuples during training: move jewel groups, provide protected-jewel cell summaries and dirty masks, and supervise reconstruction of the original field;
6. scale from four UCF classes to caption-diverse fixed-camera video, then introduce camera modeling and a thin pixel refiner only if the pure-field appearance ceiling remains binding.

A publication-grade comparison should include GaRV/VeGaS/GSVC-family reconstruction controls, pixel-space video baselines, parameter count, fitting/encoding time, memory, random-access render speed, rate-distortion, FVD or VBench-style generation metrics, edit locality, and prompt selectivity. The unique value proposition is not winning every codec metric; it is combining continuous temporal structure, persistent generation, direct manipulation, and constrained repair in one native representation. A longer-term foundation-model path would expose the coarse space-time hierarchy and event vocabulary to a multimodal transformer while retaining a continuous local generator for fine jewel marks.

# 11 Conclusion

The broad idea that Gaussian splats can reconstruct video is established prior art. The present work contributes a disciplined attempt to turn that representation into a generative and editable medium. The strongest findings are structural: anisotropic space-time support is genuinely used; effective density depends on temporal lifetime rather than raw count; near-one-jewel tokenizer locality preserves small actors; a compact hierarchy makes dense generation tractable; spatially aligned prefix context is essential for births; and local editing can preserve untouched latents exactly. These properties motivate an explicit-state alternative to opaque raster latents and a hierarchical visual event language for multimodal models. The strongest negative finding is equally important: neither semantic conditioning nor visual inpainting is solved by the present small corpus.

The project is therefore past the representation-mechanics question but before the text-to-video claim. The next milestone is unambiguous: correct prompts must beat shuffled and trained-null controls on held-out clips and during overlap-conditioned rollout, while maintaining the density, identity, color, and edit invariants already measured.

# A Reproducibility Snapshot

**Table 8:** Current reproducibility-critical constants.

| Component | Frozen or selected setting |
|---|---|
| Coordinate domain | normalized $[-1, 1]^3$ in horizontal, vertical, and time |
| Primitive feature | 22-D: center, log-covariance, P0 RGB, P1 color gradient, weight logit |
| Renderer | additive P1, learned background, kNN 64, 100M distance-pair cap |
| Dense prompt fit | 96 frames, 160-pixel short side, 16k init, 120k maximum, spatial split |
| Fine tokenizer | $32^3$ cells, 24-D local latent, sparse variable-count decode |
| Hierarchy | non-overlapping $2^3$ train-only PCA, 96 retained components, $16^3$ codes |
| Prior | rectified flow, rotating axial attention, nullable CLIP/text condition |
| Continuation | stable-ID carry plus cell-local prediction of frontier birth counts and marks |
| Editing | swept source/destination dirty mask, protected moved jewels, exact clean-cell clamp |

# B Novelty Checklist Before Submission

1. Update the literature search through the submission date, including citations that cite VeGaS, GaRV, HyperGS, and Diff4Splat.
2. Replace the provisional positioning paragraph with a feature-by-feature comparison verified against code or supplements, not abstracts alone.
3. Demonstrate text-conditioned generation from noise on held-out prompts and sources.
4. Demonstrate at least two autonomous continuation strides with no privileged future field.
5. Demonstrate a moved-object repair in which clean regions are exact and the dirty region is perceptually plausible under multiple samples.
6. Release the protocol, checkpoints, videos, and a matched baseline script.

# References


[1] Andreas Blattmann, Tim Dockhorn, Sumith Kulal, Daniel Mendelevitch, Maciej Kilian, Dominik Lorenz, Yam Levi, Zion English, Vikram Voleti, Adam Letts, Varun Jampani, and Robin Rombach. Stable video diffusion: Scaling latent video diffusion models to large datasets. *arXiv preprint arXiv:2311.15127*, 2023. URL https://arxiv.org/abs/2311.15127.

[2] Chameleon Team. Chameleon: Mixed-modal early-fusion foundation models. *arXiv preprint arXiv:2405.09818*, 2024. URL https://arxiv.org/abs/2405.09818.

[3] Hyojun Go, Byeongjun Park, Hyelin Nam, Byung-Hoon Kim, Hyungjin Chung, and Changick Kim. Videorfsplat: Direct scene-level text-to-3d gaussian splatting generation with flexible pose and multi-view joint modeling. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 26706–26717, 2025.

[4] Jonathan Ho, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi, and David J. Fleet. Video diffusion models. In *Advances in Neural Information Processing Systems*, volume 35, 2022. URL https://arxiv.org/abs/2204.03458.

[5] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkuehler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4), 2023. doi: 10.1145/3592433.

[6] Inseo Lee, Youngyoon Choi, and Joonseok Lee. Gaussianvideo: Efficient video representation and compression by gaussian splatting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2025. URL https://openaccess.thecvf.com/content/CVPR2025W/PBVS/html/Lee_GaussianVideo_Efficient_Video_Representation_and_Compression_by_Gaussian_Splatting_CVPRW_2025_paper.html.

[7] Zhan Li, Zhang Chen, Zhong Li, and Yi Xu. Spacetime gaussian feature splatting for real-time dynamic view synthesis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. URL https://openaccess.thecvf.com/content/CVPR2024/html/Li_Spacetime_Gaussian_Feature_Splatting_for_Real-Time_Dynamic_View_Synthesis_CVPR_2024_paper.html.

[8] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023. URL https://arxiv.org/abs/2210.02747.

[9] Cewu Lu, Jianping Shi, and Jiaya Jia. Abnormal event detection at 150 fps in matlab. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 2720–2727, 2013.

[10] Panwang Pan, Chenguo Lin, Chenxin Li, Jingjing Zhao, Yuchen Lin, Haopeng Li, Yunlong Lin, Kairun Wen, Yixuan Yuan, and Yadong Mu. Diff4splat: Repurposing video diffusion models for dynamic scene generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4232–4244, 2026.

[11] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139, pages 8748–8763, 2021.

[12] Barbara Roessle, Norman Mueller, Lorenzo Porzi, Samuel Rota Bulo, Peter Kontschieder, Angela Dai, and Matthias Niessner. L3dg: Latent 3d gaussian diffusion. *ACM Transactions on Graphics*, 2024. URL https://arxiv.org/abs/2410.13530.

[13] Sachin Shah, Anustup Choudhury, Guan-Ming Su, Jaclyn Pytlarz, Christopher A. Metzler, and Trisha Mittal. Gaussian representations for video. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 827–837, 2026. URL https://openaccess.thecvf.com/content/WACV2026/html/Shah_Gaussian_Representations_for_Video_WACV_2026_paper.html.

[14] Weronika Smolak-Dyżewska, Dawid Malarz, Kornel Howil, Jan Kaczmarczyk, Marcin Mazur, and Przemysław Spurek. Vegas: Video gaussian splatting. *arXiv preprint arXiv:2411.11024*, 2024. URL https://arxiv.org/abs/2411.11024.

[15] Khurram Soomro, Amir Roshan Zamir, and Mubarak Shah. Ucf101: A dataset of 101 human actions classes from videos in the wild. Technical Report CRCV-TR-12-01, University of Central Florida, 2012. URL https://arxiv.org/abs/1212.0402.

[16] Yang-Tian Sun, Yi-Hua Huang, Lin Ma, Xiaoyang Lyu, Yan-Pei Cao, and Xiaojuan Qi. Splatter a video: Video gaussian representation for versatile processing. In *Advances in Neural Information Processing Systems*, volume 37, pages 50401–50425, 2024.

[17] Longan Wang, Yuang Shi, and Wei Tsang Ooi. Gsvc: Efficient video representation and compression through 2d gaussian splatting. *arXiv preprint arXiv:2501.12060*, 2025. URL https://arxiv.org/abs/2501.12060.

[18] Xinlong Wang, Xiaosong Zhang, Zhengxiong Luo, Quan Sun, Yufeng Cui, Jinsheng Wang, Fan Zhang, et al. Emu3: Next-token prediction is all you need. *arXiv preprint arXiv:2409.18869*, 2024. URL https://arxiv.org/abs/2409.18869.

[19] Guanjun Wu, Taoran Yi, Jiemin Fang, Lingxi Xie, Xiaopeng Zhang, Wei Wei, Wenyu Liu, Qi Tian, and Xinggang Wang. 4d gaussian splatting for real-time dynamic scene rendering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 20310–20320, 2024.

[20] Bowen Zhang, Jiapeng Tang, Matthias Niessner, and Peter Wonka. Gaussiancube: A structured and explicit radiance representation for 3d generative modeling. *arXiv preprint arXiv:2403.19655*, 2024. URL https://arxiv.org/abs/2403.19655.

[21] Chunting Zhou, Lili Yu, Arun Babu, Kushal Tirumala, Michihiro Yasunaga, Leonid Shamis, Jacob Kahn, Xuezhe Ma, Luke Zettlemoyer, and Omer Levy. Transfusion: Predict the next token and diffuse images with one multi-modal model. In *International Conference on Learning Representations*, 2025. URL https://arxiv.org/abs/2408.11039.

[22] Fatimah Zohra, Chen Zhao, Shuming Liu, Yahya Al Malallah, and Bernard Ghanem. Hypergs: Fast and generalizable gaussian video representation. *arXiv preprint arXiv:2607.11500*, 2026. URL https://arxiv.org/abs/2607.11500.